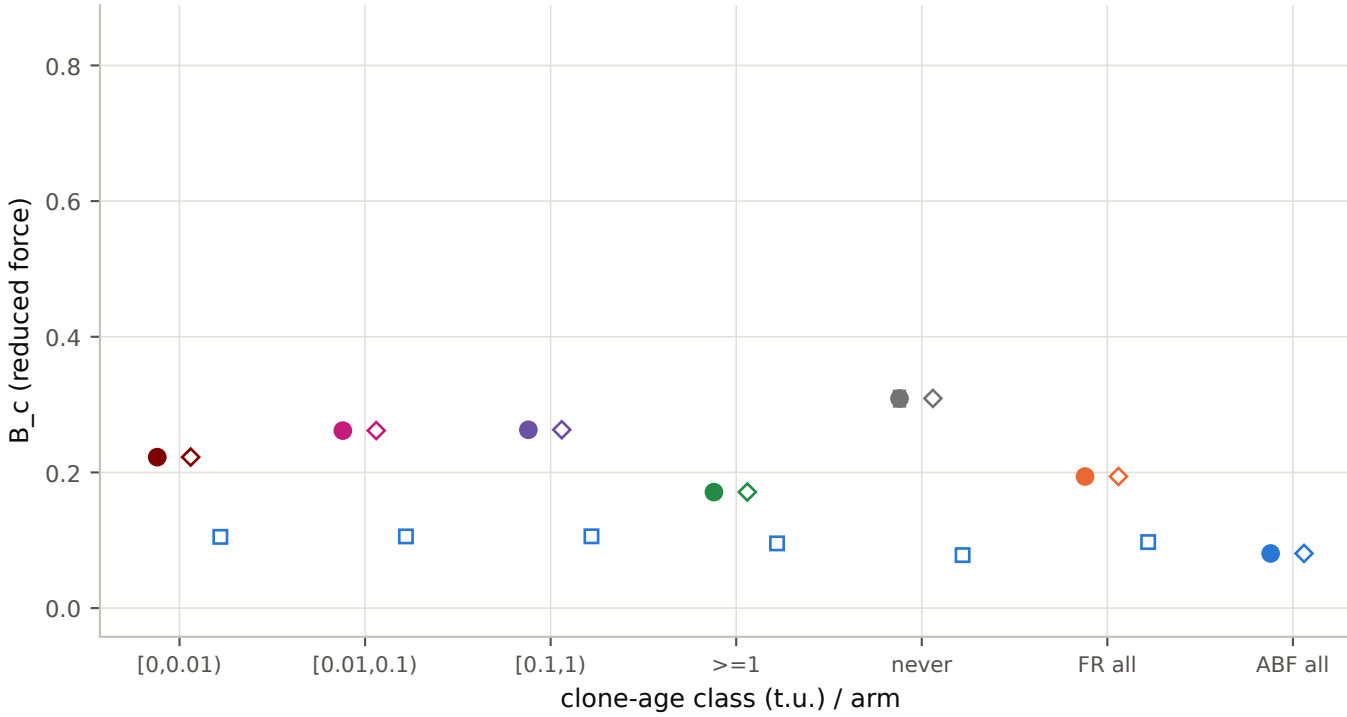
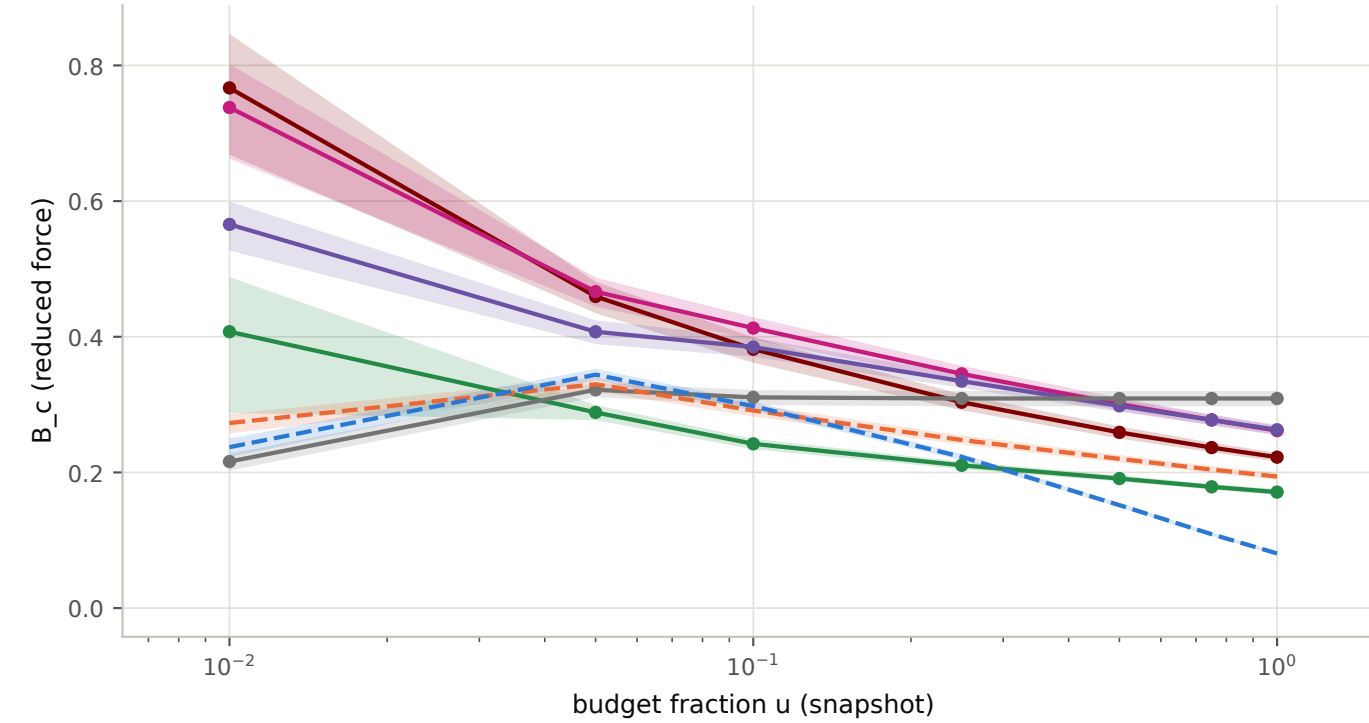


D2(ii) class-wise conditional mean-force error vs the bin-averaged F^* : I: shifted fibre, kappa = 1 (variance-matched control; constant width, curved slow fibre), N = 512
 $B_c^2 = \sum_j n_{cj} [(m_{cj} - \langle F^* \rangle_j)^2 - V_{cj}] / \sum_j n_{cj}$ over eval bins (V_{cj} : seed-cluster variance of the pooled bin mean); CIs: 10000 seed-bootstrap resamples (ABF+FR n = 32, ABF n = 32; B_c : basic interval of the debiased B_c^2). The ABF arm is the reference level. y ranges shared by every cell of this experiment at this N.

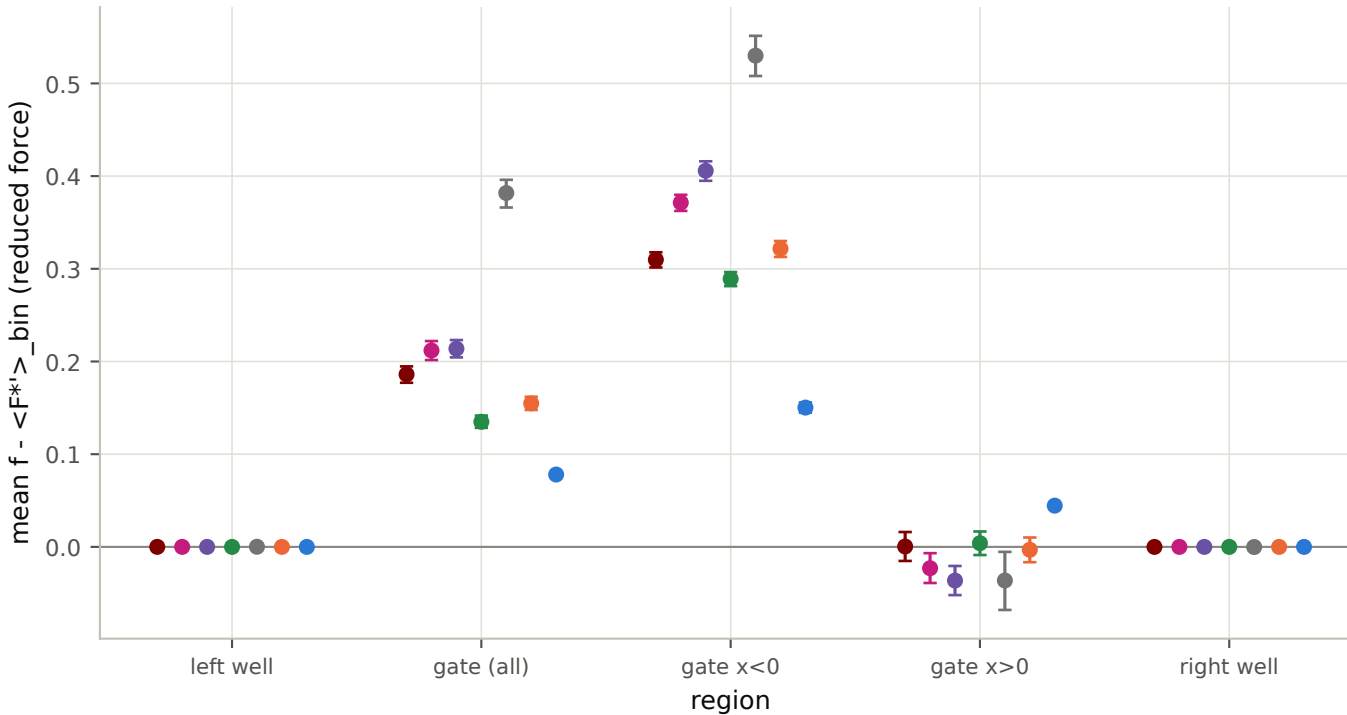
(a) debiased RMS mean-force bias B_c at the end



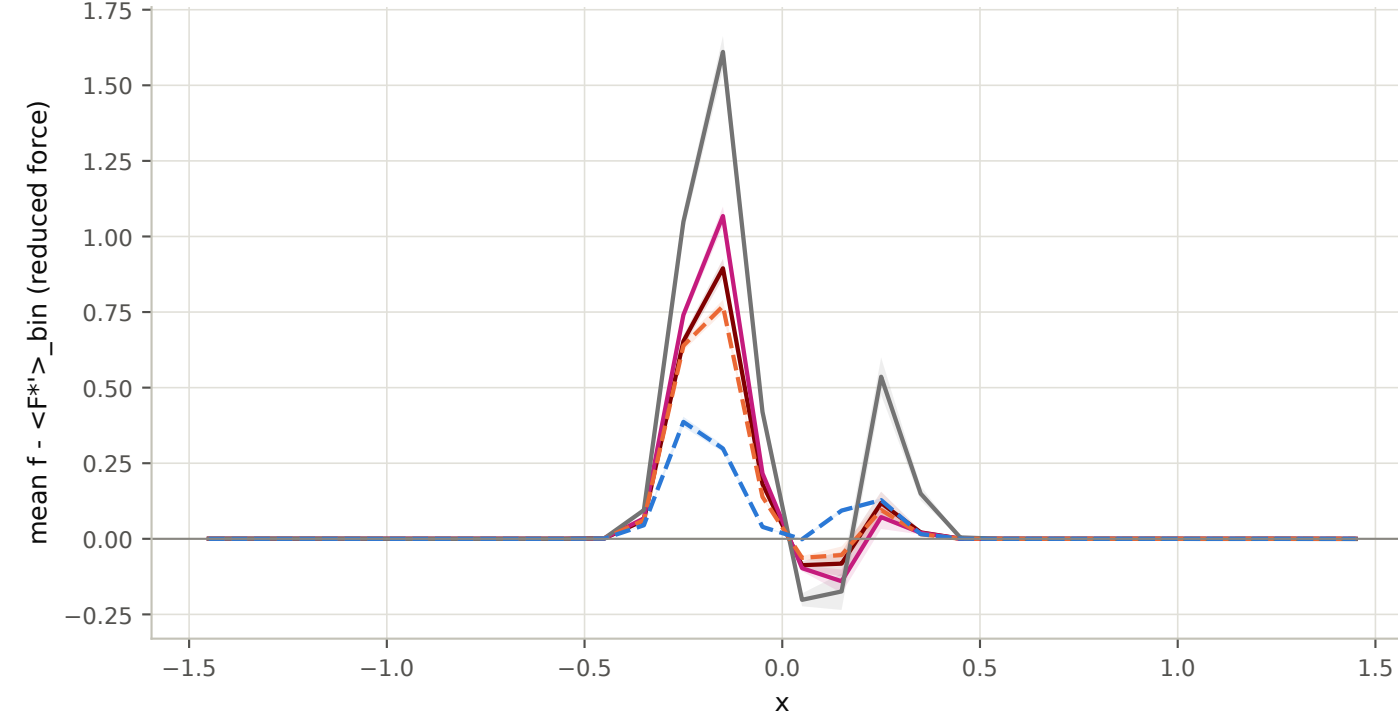
(b) B_c of the cumulative deposits at each snapshot



(c) signed deposit-weighted deviation D_{cR} at the end (eval window)



(d) deviation profile at the end (0.1-wide bins)



— clone age [0,0.01] t.u. — clone age [0.1,1] t.u. — never cloned - - ABF (all deposits = never cloned) □ (a) EXPLORATORY: ABF's B at the class's positions (matched, paired)

— clone age [0.01,0.1] t.u. — clone age ≥ 1 t.u. - - ABF+FR, all deposits ◇ raw RMS (not debiased) 95 % seed-bootstrap CI